

1)  $\int x e^{2x} dx$

$u = x$   
 $du = dx$

$dv = e^{2x} dx$   
 $v = \int e^{2x} dx = \int e^m \frac{dm}{2}$

$m = 2x$   
 $dm = 2 dx$

$v = \frac{1}{2} \int e^m = \frac{e^{2x}}{2}$

$$\begin{aligned} \int x e^{2x} dx &= x \frac{e^{2x}}{2} - \int \frac{e^{2x}}{2} dx \\ &= x \frac{e^{2x}}{2} - \frac{1}{2} \int e^{2x} dx \\ &= x \frac{e^{2x}}{2} - \frac{1}{2} \frac{e^{2x}}{2} = \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + C \end{aligned}$$

3)  $\int x \cos 5x dx = x \frac{\cos 5x}{5} - \int \frac{\cos 5x}{5} dx = x \frac{\cos 5x}{5} - \frac{1}{5} \int \cos 5x dx$

$$\begin{aligned} u &= x & dv &= \cos 5x dx \\ du &= dx & v &= \int \cos 5x dx = \int \cos m \frac{dm}{5} = \frac{1}{5} \int \cos m dm \end{aligned}$$

$$\begin{aligned} m &= 5x & v &= \frac{\sin 5x}{5} \\ dm &= 5 dx \end{aligned}$$

$$= x \frac{\cos 5x}{5} - \frac{1}{5} \frac{\cos 5x}{5} = \frac{1}{5} x \sin 5x - \frac{1}{25} \cos 5x + C$$

5)  $\int r e^{r/2} dr = r 2e^{r/2} - \int 2e^{r/2} dr = r 2e^{r/2} - 2 \int e^{r/2} dr$

$$\begin{aligned} u &= r & dv &= e^{r/2} dr \\ du &= dr & v &= \int e^{r/2} dr = \int e^m \frac{dm}{1/2} = 2 \int e^m dm \end{aligned}$$

$$\begin{aligned} m &= r/2 & v &= 2e^{r/2} \\ dm &= \frac{1}{2} dr & v &= 2e^{r/2} \end{aligned}$$

$$= r 2e^{r/2} - 2(2e^{r/2})$$

$$= r 2e^{r/2} - 4e^{r/2}$$

$$= 2(r-2)e^{r/2} + C$$

$$7) \int (x^2 + 2x) \cos x \, dx = (x^2 + 2x) \sin x - \int \sin x (2x + 2) \, dx$$

$$U = x^2 + 2x \quad dv = \cos x \, dx \quad = (x^2 + 2x) \sin x - \int 2x \sin x + 2 \sin x \, dx$$

$$du = 2x + 2 \, dx \quad v = \sin x$$

$$= (x^2 + 2x) \sin x - 2 \int x \sin x + \int 2 \sin x \, dx$$

$$= (x^2 + 2x) \sin x - 2 [-x \cos x - \int -\cos x \, dx] + 2 \cos x$$

$$U = x \quad dv = \sin x \, dx \quad = (x^2 + 2x) \sin x - 2 [-x \cos x + \sin x] + 2 \cos x$$

$$du = dx \quad v = -\cos x$$

$$= (x^2 + 2x) \sin x + 2x \cos x - 2 \sin x + 2 \cos x$$

$$= (x^2 + 2x) \sin x + (2x + 2) \cos x - 2 \sin x + C$$

$$9) \int \cos^{-1} x \, dx = x \cos^{-1} x - \int x \cdot \frac{-1}{\sqrt{1-x^2}} \, dx = x \cos^{-1} x + \int \frac{x}{\sqrt{1-x^2}} \, dx$$

$$U = \cos^{-1} x \quad dv = dx \quad = x \cos^{-1} x + \int \frac{x}{m^{1/2}} \frac{dm}{-2x} =$$

$$du = -\frac{1}{\sqrt{1-x^2}} \, dx \quad v = x$$

$$= x \cos^{-1} x - \frac{1}{2} \int m^{-1/2} \, dm$$

$$m = 1 - x^2$$

$$dm = -2x \, dx$$

$$= x \cos^{-1} x - \frac{1}{2} \left[ \frac{m^{1/2}}{1/2} \right]$$

$$= x \cos^{-1} x - m^{1/2} = x \cos^{-1} x - \sqrt{1-x^2} + C$$

$$11) \int t^4 \ln t \, dt = t^3 t \ln t - t - \int t \ln t - t \cdot 2t^2 \, dt$$

$$U = t^3$$

$$du = 3t^2 \, dt$$

$$dv = \ln t \, dt$$

$$V = \int \ln t \, dt = t \ln t - \int t \cdot \frac{1}{t} \, dt$$

$$V = t \ln t - \int \frac{t}{t} \, dt = t \ln t - t$$

$$u = \ln t$$

$$du = \frac{1}{t} \, dt$$

$$dv = dt$$

$$v = t$$

$$*13) \int t \csc^2 t \, dt = -t \cot t - \int -\cot t \, dt = -t \cot t + \int \cot t \, dt$$

$$U=t \quad dV=\csc^2 t \, dt \quad = -t \cot t + \int \frac{\cos t}{\sin t} \, dt$$

$$dU=dt \quad V=-\cot t \quad = -t \cot t + \int \frac{\cos t}{\sin t} \cdot \frac{dU}{\cos t}$$

$$U=\sin t \quad = -t \cot t + \int \frac{1}{U} \, dU = -t \cot t + \ln |\sin t| + C$$

$$dU=\cos t \, dt$$

$$15) \int (\ln^3 \sqrt{x}) \, dx = (\ln^3 \sqrt{x}) \cdot x - \int x \cdot \frac{1}{3x} \, dx = (\ln^3 \sqrt{x}) \cdot x - \int \frac{x}{3x} \, dx = (\ln^3 \sqrt{x}) \cdot x - \frac{1}{3} \int dx$$

$$U=(\ln^3 \sqrt{x}) = \ln x^{3/2} \quad dV=dx \quad = x(\ln^3 \sqrt{x}) - \frac{1}{3}x + C$$

$$dU=\frac{1}{x^{1/2}} \cdot \frac{1}{2} x^{-1/2} \, dx \quad V=x$$

$$dU=\frac{1}{3x}$$

$$77) \int \tan^{-1} 4t \, dt = \tan^{-1} 4t \cdot t - \int t \cdot \frac{4}{1+(4t)^2} \, dt = \tan^{-1} 4t \cdot t - \int \frac{m}{1+m^2} \cdot \frac{dm}{4}$$

$$U=\tan^{-1} 4t \quad dV=dt \quad m=4t \quad = t \tan^{-1} 4t - \frac{1}{4} \int \frac{m}{1+m^2} \, dm$$

$$dU=\frac{1}{1+(4t)^2} \cdot 4 \quad V=t \quad dm=4 \, dt \quad = t \tan^{-1} 4t - \frac{1}{4} \int \frac{m}{n} \cdot \frac{dn}{2m}$$

$$dU=\frac{4}{1+(4t)^2} \, dt \quad n=1+m^2 \quad = t \tan^{-1} 4t - \frac{1}{8} \int \frac{1}{n} \, dn$$

$$dn=2m \, dm \quad = t \tan^{-1} 4t - \frac{1}{8} \ln |n|$$

$$= t \tan^{-1} 4t - \frac{1}{8} \ln |1+m^2| = t \tan^{-1} 4t - \frac{1}{8} \ln |1+16t^2| + C$$

$$19) \int z^3 e^z dz = z^3 e^z - \int e^z 3z^2 dz = z^3 e^z - 3 \int z^2 e^z dz$$

$$U = z^3 \quad dv = e^z dz \quad = z^3 e^z - 3 \left[ z^2 e^z - \int e^z 2z dz \right]$$

$$du = 3z^2 dz \quad v = e^z$$

$$= z^3 e^z - 3 \left[ z^2 e^z - 2 \int z e^z dz \right]$$

$$U = z^2 \quad dv = e^z dz \quad = z^3 e^z - 3z^2 e^z + 6 \left[ z e^z - \int e^z dz \right]$$

$$du = 2z dz \quad v = e^z$$

$$U = z \quad dv = e^z dz \quad = z^3 e^z - 3z^2 e^z + 6z e^z - 6e^z + C$$

$$du = dz \quad v = e^z$$

$$21) \int \frac{x e^{2x}}{(1+2x)^2} dx = x e^{2x} - \frac{1}{2+4x} - \int \frac{1}{2+4x} e^{2x} + 2x e^{2x}$$

$$U = x e^{2x} \quad dv = \frac{1}{(1+2x)^2} dx$$

$$du = e^{2x} + 2x e^{2x} \quad V = \int \frac{1}{(1+2x)^2} dx$$

$$V = \int \frac{1}{m^2} \frac{dm}{2}$$

$$m = 1+2x \quad V = \frac{1}{2} \int \frac{1}{m^2} dm$$

$$dm = 2 dx$$

$$V = \frac{1}{2} \left( -\frac{1}{m} \right) =$$

$$V = \frac{1}{2} \left( -\frac{1}{1+2x} \right)$$

$$V = -\frac{1}{2+4x}$$

$$23) \int_0^{1/2} x \cos \pi x dx = x \frac{1}{\pi} \sin \pi x - \int \frac{1}{\pi} \sin \pi x dx = \frac{x}{\pi} \sin \pi x - \frac{1}{\pi} \int \sin \pi x dx$$

$$U = x \quad dv = \cos \pi x dx \quad n = \pi x \quad = \frac{x}{\pi} \sin \pi x - \frac{1}{\pi} \int \sin n dn$$

$$du = dx \quad V = \int \cos \pi x dx \quad dn = \pi dx$$

$$= \frac{x}{\pi} \sin \pi x - \frac{1}{2\pi} \int \sin n dn$$

$$m = \pi x \quad V = \frac{1}{\pi} \int \cos m dm \quad = \frac{x}{\pi} \sin \pi x + \frac{1}{2\pi} \cos \pi x \Big|_0^{1/2}$$

$$dm = \pi dx \quad V = \frac{1}{\pi} (\sin m)$$

$$V = \frac{1}{\pi} \sin \pi x$$

$$= 0.163957$$

$$25) \int_0^2 y \sinh y \, dy = y \cosh y - \int \cosh y \, dy = y \cosh y - \sinh y \Big|_0^2$$

$$\begin{aligned} U &= y & dv &= \sinh y \, dy \\ du &= dy & v &= \cosh y \end{aligned} \quad = \underline{2 \cosh 2 - \sinh 2}$$

$$27) \int_1^5 \frac{\ln R}{R^2} \, dR = \frac{1}{R^2} R \ln R - R - \int R \ln R \, dR$$

$$\begin{aligned} U &= \frac{1}{R^2} & dv &= \ln R \, dR \\ du &= -\frac{2}{R^3} & v &= \int \ln R \, dR \\ & & &= R \ln R - \int R \frac{1}{R} \, dR \end{aligned}$$

$$\begin{aligned} U &= \ln R & dv &= dR \\ du &= \frac{1}{R} & v &= R \end{aligned} \quad = R \ln R - R$$

$$29) \int_0^{\pi} x \sin x \cos x \, dx = x \cdot \frac{1}{2} \cos 2x - \int \frac{1}{2} \cos 2x \, dx$$

$$\begin{aligned} U &= x & dv &= \sin 2x \\ du &= dx & v &= -\frac{1}{2} \cos 2x \end{aligned} \quad = -\frac{1}{2} \left[ -\frac{x \cos 2x}{2} \right] + \frac{1}{2} \int \cos 2x \, dx$$

$$= \frac{1}{2} \left[ -\frac{x \cos 2x}{2} + \frac{1}{4} \int \cos m \, dm \right]$$

$$\begin{aligned} m &= 2x \\ dm &= 2 \, dx \end{aligned} \quad = \frac{1}{2} \left[ -\frac{x \cos 2x}{2} + \frac{1}{4} \sin 2x \right] \Big|_0^{\pi}$$

$$= \underline{-\frac{\pi}{4}}$$

$$31) \int_1^5 \frac{M}{e^M} \, dM = M \cdot \frac{1}{e^M} - \int \frac{1}{e^M} \, dM = -\frac{M}{e^M} + \int \frac{1}{e^M} \, dM = -\frac{M}{e^M} + \int \frac{1}{n} \frac{dn}{e^M}$$

$$\begin{aligned} U &= M & dv &= \frac{1}{e^M} \, dM \\ du &= dM & v &= \int e^{-M} \, dM \\ & & v &= -e^{-M} = -\frac{1}{e^M} \end{aligned} \quad = -\frac{M}{e^M} +$$

$$\begin{aligned} n &= e^M \\ dn &= e^M \, dM \end{aligned}$$

$$33) \int_0^{\pi/3} \sin x (\ln(\cos x)) dx = -(\ln(\cos x)) \cos x - \int -\cos x - \frac{\sin x}{\cos x} dx$$

$$u = \ln(\cos x)$$

$$du = \frac{-\sin x}{\cos x} dx$$

$$du = -\frac{\sin x}{\cos x} dx$$

$$dv = \sin x dx$$

$$v = -\cos x$$

$$= -(\ln(\cos x)) \cos x - \int \sin x dx$$

$$= -(\ln(\cos x)) \cos x + \cos x \Big|_0^{\pi/3}$$

$$= \frac{1}{2} (\ln 2 - \frac{1}{2})$$

$$35) \int_0^{\pi/2} \cos^{-1} x dx = x \cos^{-1} x - \int x - \frac{1}{\sqrt{1-x^2}} dx = x \cos^{-1} x + \int \frac{x}{\sqrt{1-x^2}} dx$$

$$u = \cos^{-1} x$$

$$du = -\frac{1}{\sqrt{1-x^2}} dx$$

$$dv = dx$$

$$v = x$$

$$= x \cos^{-1} x + \int \frac{x}{m^{1/2}} \frac{dm}{-2x}$$

$$= x \cos^{-1} x - \frac{1}{2} \int m^{-1/2} dm$$

$$m = 1 - x^2$$

$$dm = -2x dx$$

$$= x \cos^{-1} x - \frac{1}{2} \cdot \frac{m^{1/2}}{1/2} = x \cos^{-1} x - \sqrt{1-x^2} \Big|_0^{\pi/2}$$

$$= \frac{\pi}{6} - \frac{\sqrt{3}}{2}$$

$$37) \int e^{\sqrt{x}} dx = x e^{\sqrt{x}} - \int x \frac{e^{\sqrt{x}}}{2\sqrt{x}} dx = x e^{\sqrt{x}} - \frac{1}{2} \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$$

$$u = e^{\sqrt{x}}$$

$$du = \frac{e^{\sqrt{x}}}{2\sqrt{x}} dx$$

$$v = x$$

$$= x e^{\sqrt{x}} - \frac{1}{2} \cdot 2 \int e^m dm$$

$$= 2(x e^{\sqrt{x}} - e^{\sqrt{x}}) + C$$

$$m = \sqrt{x} = x^{1/2}$$

$$dm = \frac{1}{2} x^{-1/2}$$

$$39) \int_0^{\pi} \theta^2 \cos \theta^2 d\theta = \int_{\frac{\pi}{2}}^{\pi} m \cos m \frac{dm}{2} = \frac{1}{2} \int_{\frac{\pi}{2}}^{\pi} u \cos u du$$

$$= \frac{1}{2} [u \sin u - (-\cos u)] = [u \sin u + \cos u] \Big|_{\frac{\pi}{2}}^{\pi}$$

$$= \frac{-2 - \pi}{4}$$

$$u = \theta^2 \quad dv = \cos \theta^2 d\theta$$

$$du = 2\theta d\theta \quad v = \sin \theta^2$$

$$m = \theta^2$$

$$dm = 2\theta d\theta$$

$$u = u \quad dv = \cos u du$$

$$du = du \quad v = \sin u$$

$$41) \int x \ln(1+x) dx = \ln(1+x) \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{1+x} dx = \ln(1+x) \frac{x^2}{2} - \frac{1}{2} \int \frac{x^2}{1+x} dx$$

$$u = \ln(1+x) \quad dv = x dx$$

$$du = \frac{1}{1+x} dx \quad v = \frac{x^2}{2}$$

$$= \ln(1+x) \frac{x^2}{2} - \frac{1}{2} \int \frac{x^2}{m} dm = \ln(1+x) \frac{x^2}{2} - \frac{1}{2} \int \frac{(m-1)^2}{m} dm$$

$$= \ln(1+x) \frac{x^2}{2} - \frac{1}{2} \int \frac{m^2 - 2m + 1}{m} dm$$

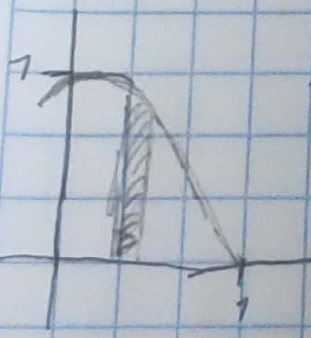
$$m = 1+x \rightarrow x = m-1$$

$$dm = dx$$

$$= \ln(1+x) \frac{x^2}{2} - \frac{1}{2} \left( m - \left( 2 + \int \frac{1}{m} \right) \right) = \ln(1+x) \frac{x^2}{2} - \frac{1}{2} \left( \frac{m^2}{2} - 2m + \ln(m) \right)$$

$$= \ln(1+x) \frac{x^2}{2} - \frac{1}{2} \left[ \frac{(1+x)^2}{2} - 2(1+x) + \ln(1+x) \right]$$

$$61) y = \cos\left(\frac{\pi x}{2}\right), y=0, 0 \leq x \leq 1, \text{ rev}$$



$$\int_0^1 \cos\left(\frac{\pi x}{2}\right) dx \int \cos(m) dm \quad \left( \frac{2\pi}{\pi} = 2\pi \int_0^1 x \cos\left(\frac{\pi x}{2}\right) dx \right)$$

$$= 2\pi \left[ x \cdot \frac{2}{\pi} \sin\left(\frac{\pi x}{2}\right) - \int \frac{2}{\pi} \sin\left(\frac{\pi x}{2}\right) dx \right]$$

$$= 2\pi \left[ \frac{2x}{\pi} \sin\left(\frac{\pi x}{2}\right) - \frac{2}{\pi} \cdot \frac{\pi}{2} \left( -\cos m \right) \right] = \frac{4}{\pi^2} \cos\left(\frac{\pi x}{2}\right)$$

$$u = x \quad dv = \cos m$$

$$du = dx \quad v = \frac{2}{\pi} \sin\left(\frac{\pi x}{2}\right)$$

$$2\pi \left[ \frac{2x}{\pi} \sin(m) + \frac{4}{\pi^2} \cos m \right] \Big|_0^{\pi/2}$$

$$= \frac{8}{\pi} \left[ \frac{\pi}{2} - 1 \right] = \frac{4(\pi - 2)}{\pi}$$

(b3)  $y = e^{-x}$ ,  $y = 0$ ,  $x = -1$ ,  $x = 0$ ,  $x = 1$   
 $e^{-x} = -1$ ,  $y = \frac{1}{e^x}$   $e^x = \frac{1}{y}$   $x = \frac{\ln 1}{y}$

$$x = \ln 1 - \ln y$$

$$x = -\ln y$$

$$-\ln = -1$$

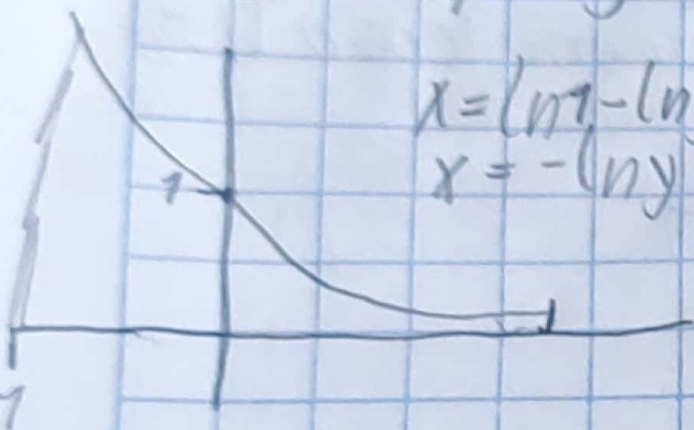
$$\ln y = 1$$

$$y = e$$

$$V = 2\pi r h dx$$

$$r = 1 - x$$

$$h = e^{-x}$$



$$V = \int_{-1}^0 2\pi(1-x)e^{-x} dx = 2\pi \int_{-1}^0 e^{-x} dx \int_{-1}^0 x e^{-x} dx =$$

$$u = x$$

$$du = dx$$

$$dv = e^{-x}$$

$$v = -e^{-x}$$

$$= -x e^{-x} - \int e^{-x} dx$$

$$2\pi [-(-e^{-x} - x e^{-x} - e^{-x})]_{-1}^0$$

$$= \underline{\underline{7.08}}$$